Lab 6: Code and Branch Coverage Analysis

Pizza Order Application | PES University

Repository: SETAPESU26/L6_L7_24_pizza_order
Application under test: pizza_order.py
Testing tools: Python 3.11.2, pytest 9.1.1, coverage.py 7.1.0

1. Objective

The objective of this lab is to understand and measure code coverage and branch coverage, identify untested parts of a program, and improve test completeness by adding test cases for normal inputs, invalid inputs, boundary conditions and alternate decision outcomes. The lab handout asks for at least 90% code coverage and at least 90% branch coverage.

2. Application Overview

The pizza order application contains three functions:

Function	Purpose
pizza_price(size, toppings, is_cheese_burst)	Calculates pizza price using size, number of toppings and cheese-burst selection. It validates s
order_discount(order_total, day, is_student)	Calculates a discount based on the order amount, day of the week and student status.
delivery_status(minutes_taken, distance_km)	Returns the delivery outcome based on delivery distance and time.

3. Initial Test Results

The starter test file contained two tests: one for a plain medium pizza and one for an on-time delivery. Both tests passed, but the initial report showed only 41% statement coverage for pizza_order.py and 34% branch coverage for pizza_order.py. This indicated that many application statements and decision outcomes were not exercised by the original tests.

4. Code Coverage Analysis

A separate test file, CodeCoverageTest_pizza_order.py, was created. It contains 20 test functions covering pizza sizes, topping calculations, cheese-burst conditions, invalid inputs, discount scenarios and delivery outcomes. An incorrect expected value in the five-topping large pizza test was corrected from 670 to 680 after comparing it with the application's pricing calculation.

Command used:

```
coverage erase
coverage run -m pytest -v CodeCoverageTest_pizza_order.py
coverage report -m
```

Result: 20 tests passed. The application file pizza_order.py had 34 statements, zero missing statements and 100% coverage. The combined report also showed 100% total coverage.

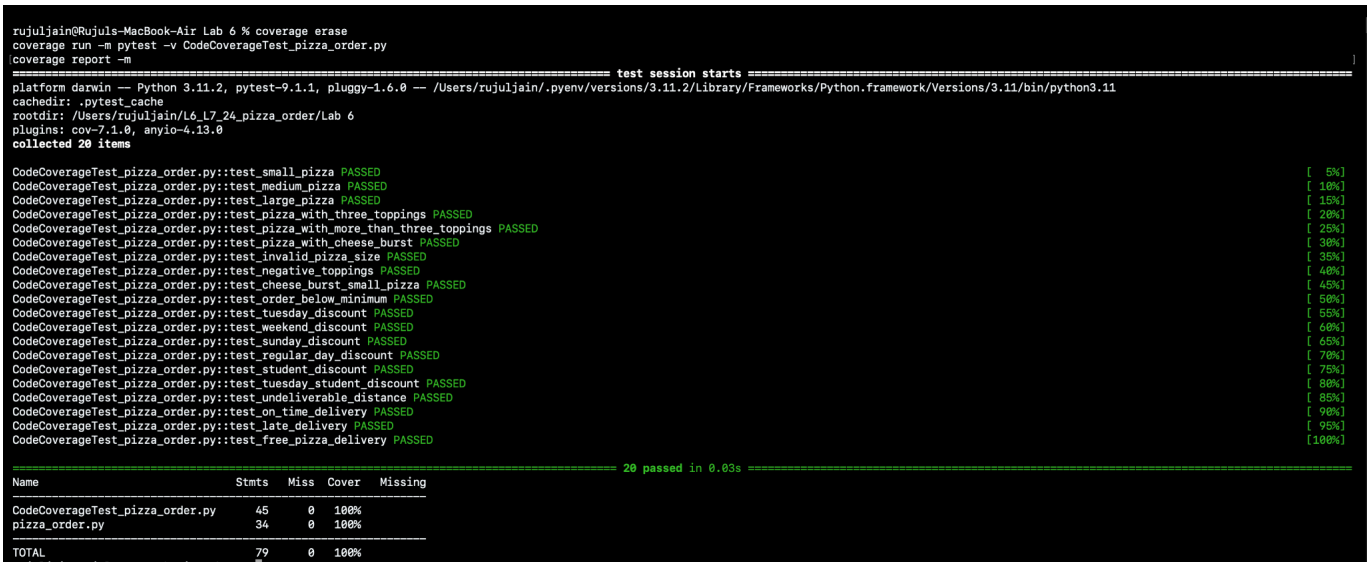


Figure 1. Terminal output showing all 20 code coverage tests passing and 100% statement coverage.

5. Branch Coverage Analysis

A second, separate test file, BranchCoverageTest_pizza_order.py, was created to exercise both outcomes of the application's conditional decisions. The tests cover valid and invalid pizza sizes, negative and non-negative toppings, topping-count pricing paths, cheese-burst availability, minimum order amount, Tuesday/weekend/regular-day discounts, student discount addition, distance eligibility and the delivery-time outcomes.

Command used:

```
coverage erase
coverage run --branch -m pytest -v BranchCoverageTest_pizza_order.py
coverage report -m
```

Result: 17 tests passed. The report for pizza_order.py showed 34 statements, zero missing statements, 24 branches, zero partial branches and 100% coverage. The total report also showed 100% coverage.

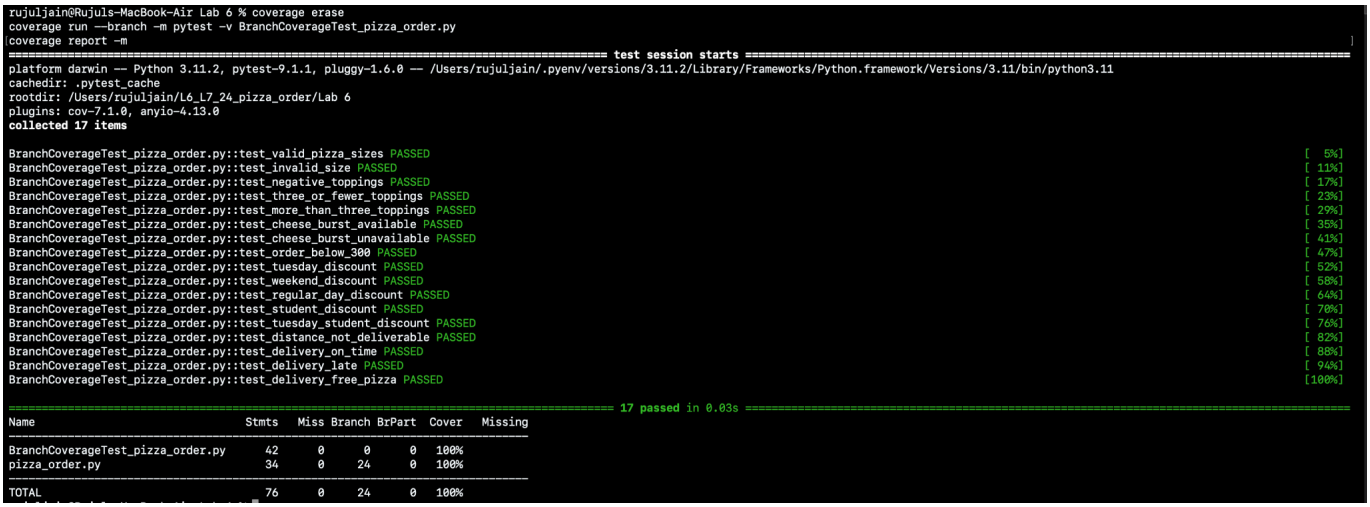


Figure 2. Terminal output showing all 17 branch coverage tests passing, with 24 branches and no partial branches.

6. Results Summary

Measure	Initial result	Final result
Statement coverage (pizza_order.py)	41%	100%
Branch coverage (pizza_order.py)	34%	100%
Code coverage test cases	2 starter tests	20 passed
Branch coverage test cases	2 starter tests	17 passed

7. Conclusion

The initial tests exercised only a small part of the pizza order application. Additional tests were written to cover normal cases, invalid inputs, boundary conditions and alternate decision outcomes. The final recorded runs achieved 100% statement coverage and 100% branch coverage for pizza_order.py, exceeding the minimum 90% target specified in the lab handout. These results indicate that every measured statement and branch in the application was exercised by the respective test runs.

8. Repository Deliverables and Commit Status

The two test files are included separately in this deliverables package, along with the terminal screenshots used as evidence. The first Git commit was confirmed in the working session: 24dee04 — “Add tests for 100 percent code coverage”. The second branch-coverage commit had not yet been confirmed at the time this report was prepared. After adding the branch test file and both screenshots to the repository, create/verify the second commit and check the repository status before submission.

Files included:
• CodeCoverageTest_pizza_order.py
• BranchCoverageTest_pizza_order.py
• Code_Coverage_Result.png
• Branch_Coverage_Result.png
• This report (PDF)